decentralised data collection approach is followed as healthcare workers often have to be in areas with no data or mobile tower connectivity.

Records are synced to the cloud and displayed on the analytics dashboard. Central health control offices can access these insights, and reach out to the people in need and act accordingly. They can better schedule immunisation drives, manage resources, predict demands and engage with communities using dialect-specific, voice-based reminders.

Currently, the NGO is working in rural Rajasthan, covering more than 4000 infants annually with vaccination outreach programmes. It is following the decentralised team-based delivery model in each village for 20 days a month.

Currently, necklaces are given to infants to track child immunisation records in the first two years after birth. Going ahead, the NGO will also start tracking maternal and child health, and the pendants will be first worn by the expecting mother before being passed on to the child to link their data.

2 AUGMENTED REALITY ANYWHERE Using A Smartphone

Wherever our real world syncs with the digital, our lives become easier. For this reason, technopreneurs are designing augmented reality (AR)- and virtual reality (VR)-based solutions that can be used anywhere, be it at homes, offices, industries or medical facilities. Tesseract Inc.—led by Massachusettes Institute of Technology alumni Kshitij Marwah—has developed its own AR platform, called Holoboard, which utilises the user's smartphone.

Holoboard consists of an AR headset that can be connected to a smartphone (condition to compatibility) to create the augmented simulation. The user can simply slide the smartphone inside the holder of the headset to get going. The system utilises the camera and sensors of the smartphone in sync with its own sensory components to operate. Essentially, it creates a virtual AR display, which is amplified in front of the eyes at a maximum resolution of 3840 × 2160 pixels.

Configuration and content streaming are done through a proprietary mobile application (for Android and iOS). Display is created by an off-axis, curved, parabolic, semi-reflective optic lens that is modelled after bug eyes. It creates an 80-degrees field-of-view.

Holoboard uses an inertial movement unit (IMU) sensor with low latency and the marker-based tracker of the smartphone's camera to track the user's movements and position the display accordingly. It works at a rate of 30 frames per second and has three degrees of freedom (DoFs).

Holoboard can be availed in three models, each having its own USP.

- The basic model is for home and regular use, like watching movies, gaming, remotely connecting with people and so on.
- The premium model can be used more interactively through gestures to get work done quickly.
- The enterprise model can be used by businesses, design houses and workstations to create designs virtually and to collaborate with team members.

In basic mode, AR can be operated using an interactive Bluetooth remote controller with programmable buttons and a touchpad to sense gestures (tap, rotate, grab, etc).

In premium mode, the user can carry out the controls with hand gestures from any position within an intuitive virtual sphere created surrounding the headset. This is done with the help of an inbuilt hand-tracking module and an add-on near-field infrared projection accessory that can help manipulate the holograms.

The enterprise mode offers 6-DoF interaction with a SLAM tracking module that enables envi-